

ThreatPulse CTI Operations Manual

Standard Operating Procedure (SOP) & Analytical Training Guide for Threat Intel & SOC Teams

Target Audience: Threat Intel Analysts, SOC Lead Engineers, CISO Staff

Ingestion Feeds: 30+ Tier-1 Intel Sources

Classification: RESTRICTED // TLP:AMBER+STRICT

Enrichment Engine: FIRST.org EPSS + NVD CVSS v3.1

1. System Overview & Architecture

ThreatPulse is a high-signal Cyber Threat Intelligence (CTI) and vulnerability triage dashboard. Built for distraction-free security operations, it aggregates, normalizes, deduplicates, and enriches intelligence from over **30+ tier-one feeds** in real time into 3 distinct stream categories:

- **Active Threat Intel & CVEs:** Reserved for high-urgency operational items: zero-days, CISA KEV disclosures, RCE advisories, ransomware campaigns, and CVE patches.
- **Security Advisories:** Platform patch bulletins and OS updates (Debian, Ubuntu, AWS, FreeBSD, Kubernetes, Netgate pfSense).
- **Informational & Research:** Tech journalism, industry analysis, and corporate security commentary (The Register, The Hacker News, Dark Reading, SANS ISC, Cloudflare Blog).
- **IOC Watch Engine:** Deduplicates atomic indicators (hashes, URLs, domains, IPs) with self-contained corroboration and decay modeling.

2. Scoring Models & Mathematical Formulas

ThreatPulse utilizes two transparent mathematical scoring algorithms to prioritize threat advisories and evaluate atomic indicator confidence.

A. Priority Scoring Algorithm (0 - 110+ Points)

Advisories are ranked in **Urgent** sort mode and the **Top Priority Spotlight** hero widget using a composite risk score:

Component	Weight / Points	Analytical Purpose
EPSS Score	Up to +40 pts (score × 40)	Direct measure of real-world exploitation probability in the wild (0.0–1.0 scale).
CVSS v3.1 Base Score	Up to +30 pts (score × 3)	Technical severity impact rating from NIST NVD (0.0–10.0 scale).
CISA KEV Proximity	+15 to +35 pts	+35 pts if due in ≤3 days / overdue; +20 pts if due in ≤7 days; +15 pts for active KEV.
Critical Severity Tier	+15 pts	Awarded for explicit zero-days, active RCEs, or confirmed ransomware campaigns.

B. IOC Watch Confidence Model (0 - 100% Scale)

Atomic indicators in the **IOC Watch** modal receive a confidence rating (High / Medium / Low) calculated at ingestion:

Indicator Class	Base Confidence	Analytical Justification
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SHA-256 Hash	75% Base	Byte-exact file signature with near-zero false positive rate.
SHA-1 / MD5 Hash	72% / 68% Base	File hashes; MD5 weighted slightly lower due to collision potential.
URL	55% Base	Path-level threat indicator; specific to payload delivery or C2 endpoints.
Domain / Hostname	40% Base	Fully qualified domain; subject to sinkholing or domain recycling.
IPv4 / IPv6 Address	25% Base	Volatile infrastructure; highly prone to proxy/CDN sharing and IP recycling.

- **Multi-Source Corroboration Bonus (+15% per additional feed):** When an indicator is reported by multiple independent feeds (e.g. ThreatFox + URLhaus), its confidence increases by +15% per additional source.
- **CDN / Cloud Allowlist Suppression:** IPs falling inside Cloudflare, AWS CloudFront, GCP, or Azure CIDRs are hard-suppressed to **5% Low Confidence** to eliminate shared-hosting false positives.
- **7-Day Freshness Decay & Purge:** Unseen IP/domain indicators decay over 7 days and are automatically purged to prevent stale IOC clutter.

3. Analyst Feature Sitemap & Operating Controls

ThreatPulse provides dedicated operational controls designed to streamline CTI shift handovers, daily triage, and executive reporting.

Feature / Control	Location / Shortcut	Operational Purpose
Triage Briefing (24h)	Toolbar Slider	Filters feed to items published in the last 24h + active CISA KEVs. Default mode for morning shift handovers.
Command Palette	Ctrl+K / Cmd+K	Instant fuzzy-search across all advisories by CVE ID, threat actor, malware family, vendor, or title.
Executive Digest Export	Toolbar ■ Digest	Copies a formatted Markdown executive summary of today's top Critical/High items ready to paste into Slack/Teams.
IOC Watch Modal	Header ■ IOC Watch	Structured table of deduplicated indicators. Supports interactive column sorting (Type, Value, Family, Confidence, Date, Sources).
SIEM / EDR CSV Export	IOC Modal ■ Export CSV	Exports filtered IOC lists directly into CSV format for ingestion into Splunk, CrowdStrike, Sentinel, or firewalls.
KEV Ticker Navigation	Hero Card [← →]	Cycles through unread critical advisories. Manual arrow buttons pause auto-rotation so analysts can finish reading.
Tag & Entity Filters	Toolbar ■■ Filters	Checkbox grid filtering by OS (Windows/Linux), Cloud (AWS/Azure), Vendor, or Threat Type (Ransomware/RCE).

4. Analyst Daily SOP Checklist

To maximize threat coverage and ensure no critical vulnerability is missed during daily operations, follow this 5-step workflow:

- **Step 1 — Morning Shift Handover (08:00):** Ensure *Today's Triage Briefing (24h)* is active. Review the **Executive Highlights** → **KEV Ticker** for any unread CISA Known Exploited Vulnerabilities published overnight.
- **Step 2 — High-Urgency Sorting:** Toggle the sort control from **Newest** to **Urgent**. Triage the top items in **Active Threat Intel & CVEs** carrying high EPSS (>50%) or CVSS (>8.0) scores.
- **Step 3 — Executive Reporting:** Click ■ Digest on the toolbar to generate the daily Markdown summary. Paste into the SOC channel and executive briefing thread.
- **Step 4 — Blocklist Enrichment (12:00 & 16:00):** Open ■ IOC Watch. Filter by *High Confidence* or *Hashes/IPs*. Click Export CSV to generate blocklists for SOC perimeter firewalls and EDR agents.
- **Step 5 — Deep Search & Bookmarking:** Use Ctrl+K to jump directly to vendor-specific advisories (e.g. 'Fortinet', 'Palo Alto', or 'CVE-2026-...'). Click ★ Save on critical items to track them in the Saved queue.

Analyst Efficiency Tip: ThreatPulse static payloads auto-refresh in the background every 5 minutes. You can leave the dashboard open on secondary SOC monitors without causing browser memory leaks or extra server overhead.